

ASSIGNMENT DETAILS

Unit Code	COS10025	Unit Title	
Tutorial/Lab Group	GT05	Lecturer/Tutor Name	
Assignment Title	Project reflection report (individual)		
Due date	09/12/2024	Date Received	11/11/2024

DECLARATION

For both individual and group assignments, in the case of assignment submission on behalf of another student, it is assumed that permission has been given. The University takes no responsibility for any loss, damage, theft, or alteration of the assignment.

To be completed if this is an individual assignment

I declare that this assignment is my individual work. I have not worked collaboratively, nor have I copied from any other student's work or from any other source/s, except where due acknowledgment is made explicitly in the text, nor has any part been written for me by another person.

Student Details	Student ID Number	Student Name	Student Signature
Student 1			

To be completed if it

We declare that this is a group assignment and that no part of this submission has been copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part been written for us by another person.

Student Details	Student ID Number(s)	Student Name(s)	Student Signature (s)
Student 1			
Student 2			
Student 3			
Student 4			
Student 5			

MARKER'S COMMENTS

Total Mark	Marker's Signature	Date
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EXTENSION CERTIFICATE

This assignment has been given an extension by

Unit Convenor	
Extended due date	Date Received

COS10025 TECHNOLOGY IN AN INDIGENOUS CONTEXT PROJECT

Final project reflection report

Bachelor of Computer Science

**Project Title: Use of Emerging Technologies to Solve
Remote Indigenous Communities Challenges**



School of Engineering, Computing and Science
Swinburne University of Technology
Sarawak Campus

9 December 2024

Declaration

I hereby declare that this report is the result of my project work except for quotations and citations which have been duly acknowledged. I also declare that it has not been previously or concurrently submitted for any other degree at Swinburne University of Technology Sarawak Campus.

Student Name:

Student ID:

Date of Submission: 9/12/2024

Signature:

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SAMPLE

This section should be 1/2 page. This section outlines the actual project, design for a remote Indigenous township. It should be about half a page. It should introduce the problem, the township

Part A: Introduction

The introduction describes the project clearly and concisely with background/motivation. The township is clearly detailed. Citations/references are included.

This project focuses on providing solutions using emerging technologies to solve the problems faced by the indigenous community in a particular remote area of Malaysia. The following sections include a detailed project description and a recommended design idea aimed at tackling the identified problems.

1.1 Project Description

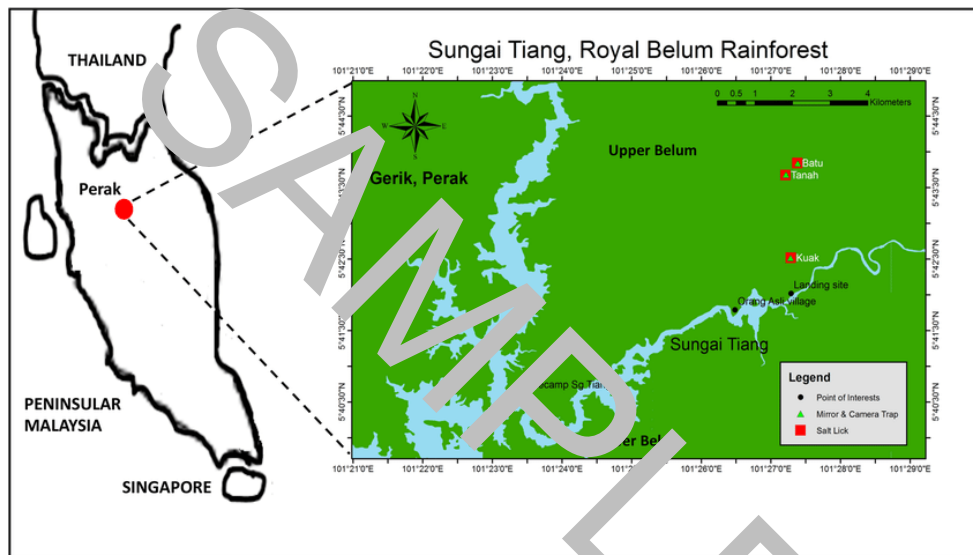


Figure 1. Location of Kampung Sungai Tiang (Mahmud et al. 2020)

include more in-text citation at part A

The team has decided to focus on providing solutions to resolve the clean water access challenges faced by the Orang Asli Jahai tribe in Kampung Sungai Tiang. The village is situated in Royal Belum State Park, located within the Belum-Temengor forest complex in the northern part of Perak (Schwabe et al. 2015, p. 56). This area is one of the world's oldest undisturbed primary forests, rich in flora and fauna, including large mammals such as Malayan tiger, Sumatran rhinoceros and Asian elephants (Schwabe et al. 2015, p. 57). The village is accessible only by boat from Tasik Banding jetty (Mahmud, Shaffii & Yunus 2022, p. 2). The terrain is relatively flat with some sloping areas. The community consists of approximately 400 people across 60 families (Mahmud, Shaffii & Yunus 2022, p. 2). Their houses are elevated and constructed mostly from wood, with some built with bamboo. Due

to its remote location, the area lacks 24 hours electricity and internet connectivity (Mahmud, Shaffii & Yunus 2022, pp. 4-5).

The community relies heavily on the nearby river and rainwater as primary water supply (Mahmud, Shaffii & Yunus 2022, p. 2). However, research shows that the water quality of Sungai Tiang falls under Class III of the National Water Quality Standards for Malaysia, which indicates the need for extensive water treatment (The DOE Environmental Quality Report, cited in Mohamad et al. 2024, pp. 106-107). Critical water quality parameters which require attention are pH value and Biochemical Oxygen Demand (BOD) levels. According to Mohamad et al.'s (2024, p. 108), the pH level of Sungai Tiang is 6.22, which falls below the safe range for human contact (6.5-8.5). Additionally, Kamaruddin et al. (2023, p. 313) reported high levels of microbial contamination in Sungai Tiang, identifying the presence of *E. coli* and *Salmonella spp.*, which pose annual infection risks of up to 22.9% and 45% respectively. This aligns with Mohamad et al.'s (2024, p. 113) finding that the BOD level in Sungai Tiang is 11.64 mg/L. These factors demonstrate that the water is polluted and poses significant health risk to the community. Therefore, a solar-powered water purification system is proposed to address microbial contamination and correct pH imbalances.

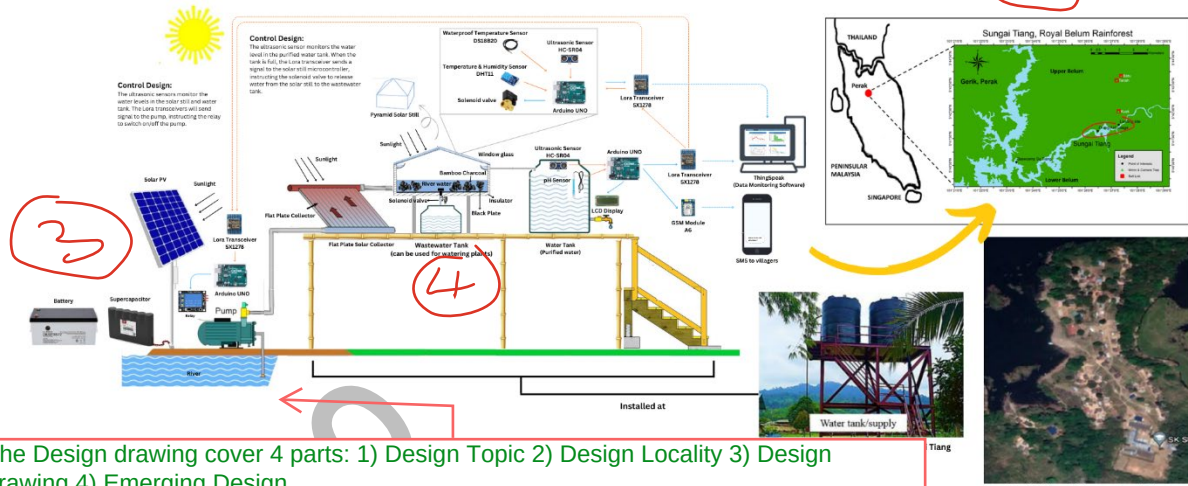
This section should be 1 page. In this section, choose 1 option for further development. Discuss the benefits, and opportunities for future use, in particular focusing on the suitability of the chosen design idea for the community and location.

1.2 Recommended Option to Proceed

①

Hybrid Solar Still and IoT-Integrated Water Purification System

②



The Design drawing cover 4 parts: 1) Design Topic 2) Design Locality 3) Design drawing 4) Emerging Design

Figure 2. Hybrid Solar Still and IoT-Integrated Water Purification System Design Drawing

Provides important details highlighted for a recommended design idea. Discussed shared value, benefits, and opportunities for a recommended solution.

The recommended design is a hybrid solar still water purification system integrated with IoT devices for resolving the insufficient clean water supply issue in Kampung Sungai Tiang (Mahmud, Shaffii & Yunus 2022, p. 1). Solar collector and solar still, the main components for purifying water in the design, work together to purify water through evaporation process. This process removes impurities such as dissolved solids and harmful microorganisms. Additionally, bamboo charcoals in the solar still enhances the removal of iron particles, further improving the water quality (Baruah et al. 2011, p. 457).

Automation is achieved through the cooperation of ultrasonic sensors installed in the tanks monitoring the water levels as well as relay module and solenoid valve controlling the water flow, reducing the need for manual intervention. To ensure long-term usability, the IoT sensors monitor key parameters such as water temperature, still temperature and pH level and sends the data to ThingSpeak (Benghanem, Mellit & Emad 2022, p. 3035) via LoRa technology for performance analysis and failure predictions.

The usage of solar energy for the entire purification process ensures that the system can operate effectively in Kampung Sungai Tiang, which lacks 24-hours electricity (Mahmud, Shaffii & Yunus 2022, p. 1). To further enhance the efficiency, supercapacitors and batteries are added to ensure that the system can also work during nighttime or unfavourable

weather conditions, maximising the availability of purified water for the community. Furthermore, real-time SMS updates on the purified water conditions are provided to the community, ensuring they stay informed about the availability of purified water (Benghanem et al. 2021, p. 5).

The following sections will explain the benefits, opportunities for future use, and the suitability of this design for Kampung Sungai Tiang in greater detail.

1.2.1 Design Benefits

The solar still utilizes sunlight to purify the water through evaporation process. Based on the study conducted by Downing and Blunt (cited in Ahmad, Khan & Farid 2024, p. 330), sunlight exposure for a few hours has been proven to kill the bacteria present in water. Additionally, research by Abujazar et al.'s (2018, p. 93) demonstrated the high efficiency of solar still in improving most water quality parameters, especially Total Dissolved Solids (TDS), Total Suspended Solids (TSS) and conductivity, proving that the purified water produced by solar still aligns with the Malaysia's drinking water quality standard (Abujazar et al. 2018, p. 94). This ensures that the community in Kampung Sungai Tiang are provided with clean water supply, thus preventing the community from waterborne diseases.

The combination of automation through IoT devices and LoRa technology, along with the use of solar energy, results in low operational costs, which is suitable for the community in long run. With no need to pay for labour or electricity charges, the system offers a cost-effective solution to ensuring continuous access to clean water for the community.

Furthermore, the components used in the system have minimal impact to the environment. According to Al-Madhhachi and Smaisim (2021, p. 312), solar stills emit insignificant amount of harmful substances, such as sulphur dioxide and phosphate, over their 20 years lifespan. The water tanks used in the design are made from polypropylene (PP) and high-density polyethylene (HDPE) (Jayapalan & Vijay 2022, p. 1). These materials are recyclable through pyrolysis, a recycling method which effectively reduces environmental issues (Achilias et al. 2007, p. 542).

1.2.2 Opportunities for Future Use

The hybrid solar still water purification system has significant potential for broader applications in the future. By scaling up the system to include multiple solar stills, the system can provide clean water to larger communities. This makes it suitable for regions with high clean water supply demand or growing populations.

Moreover, the integration of IoT devices and LoRa technology introduces various possibilities for the system's future applications in different regions. The ability to automatically control the operations and monitor the performance makes it suitable for deployment in other remote or off-grid areas as well, which often face issues of limited internet connectivity. This approach ensures the system can maintain its functionality with minimal maintenance, which is a crucial feature for locations that are difficult to access.

1.2.3 Suitability

The solar purification system purely operates on sunlight, making it an ideal solution for Kampung Sungai Tiang, where 24-hour electricity is unavailable. With the inclusion of supercapacitors and batteries for solar energy storage, the community is assured of a continuous supply of clean water even during adverse weather conditions. By relying on clean and renewable energy as the main power source, the system promotes long-term environmental sustainability. Studies have shown that components such as solar stills (Jasim, Ahmed & Alaiwi 2023, p. 98) and solar PV panels (Sindhuja, Abu-Al Ez & Kahn 2021, p. 1) can last for up to 30 years with minimal maintenance. This durability reduces the need of frequent replacement of broken components, further ensuring environmental health over prolonged use. Moreover, the automated design of the system eradicates the need for technical expertise, making it user-friendly and easy for the community to adopt. The system is also designed to be installed in an existing elevated area used for water tanks. This placement not only prevents animal interference, but most importantly it does not affect the original lifestyle of the indigenous community, ensuring its cultural acceptance.

[This section is 1-2 pages. In this section, refer to events, activities (week 2 – week 11 group activities) and people specific to your group (i.e., avoid the use of 'we/he/she' in this discussion, instead refer to your group members by name, do not list general guides/recommendations for group work) & (add evidences from weekly workshop team activities, weekly seminar reflections or assessments)

Part B: Project Reflection

The summary of a group reflection is well-articulated with the group work strategies, and challenges with team members. Clearly described with appropriate events/actions in regard to team organisation, meetings, and project delivery.

This section reflects my experience during the process of completing the current project.

2.1 Group Work Reflection

My experience working in a group with throughout the semester is discussed in the following subsections.

Describe the group work strategies/processes that worked for your team

2.1.1 Group Work Strategies/Processes that Worked

During the Week 2 Workshop and I formed a team to complete this project. After discussion, Sin Mim was chosen as the team leader. To facilitate communication, we shared our contact information and Sin Mim created a WhatsApp group (see Figure 3) which became our primary communication platform.

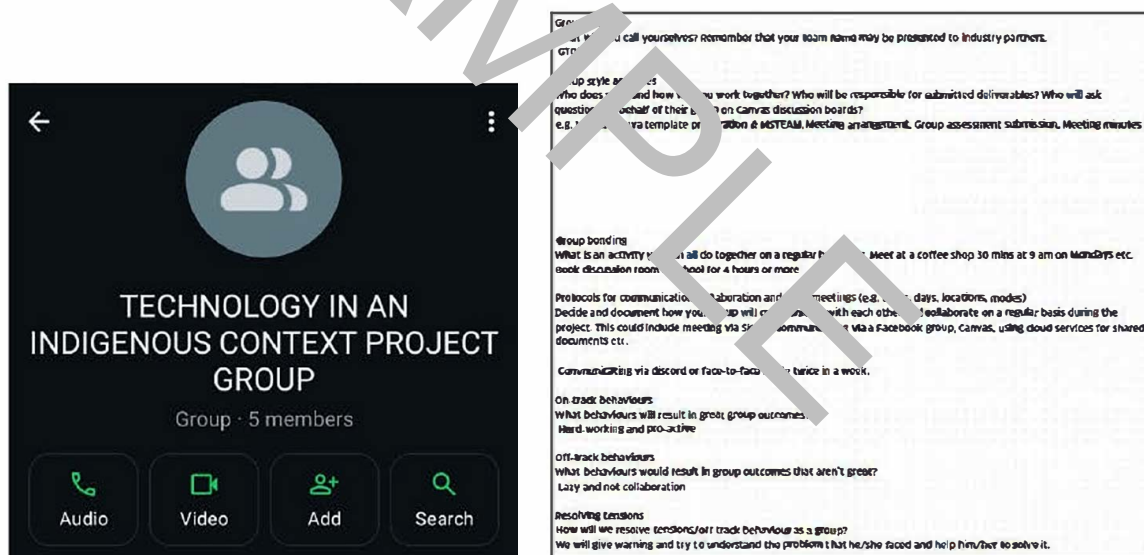


Figure 3. Formation of WhatsApp Group & Role Assignment for Each Member during Week 2 Workshop

Each team member was assigned specific responsibilities by Sin Mim (see Figure 3). My role was to arrange and announce meeting schedules, Ellie was assigned to create Canva templates for workshops, Aryn was responsible for the group work submissions, and Cyndia was responsible for documenting the meeting minutes. To keep the project on track, we agreed to hold meetings twice a week, in addition to the tutorial sessions.

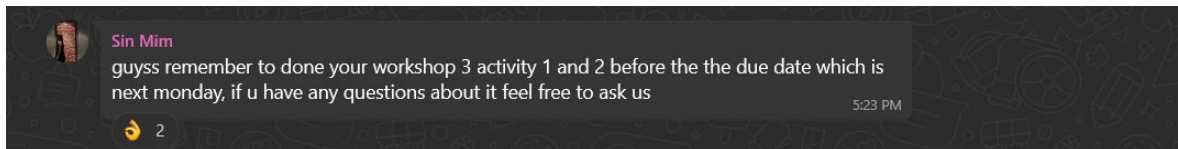


Figure 4. Reminder from Sin Mim on Workshop Submission Due

Each member fulfilled their responsibilities effectively. Sin Mim, as the team leader, consistently reminded everyone to complete their tasks the day before the submission deadline (see Figure 4). This ensures that the works were submitted on time without last-minute rushing.

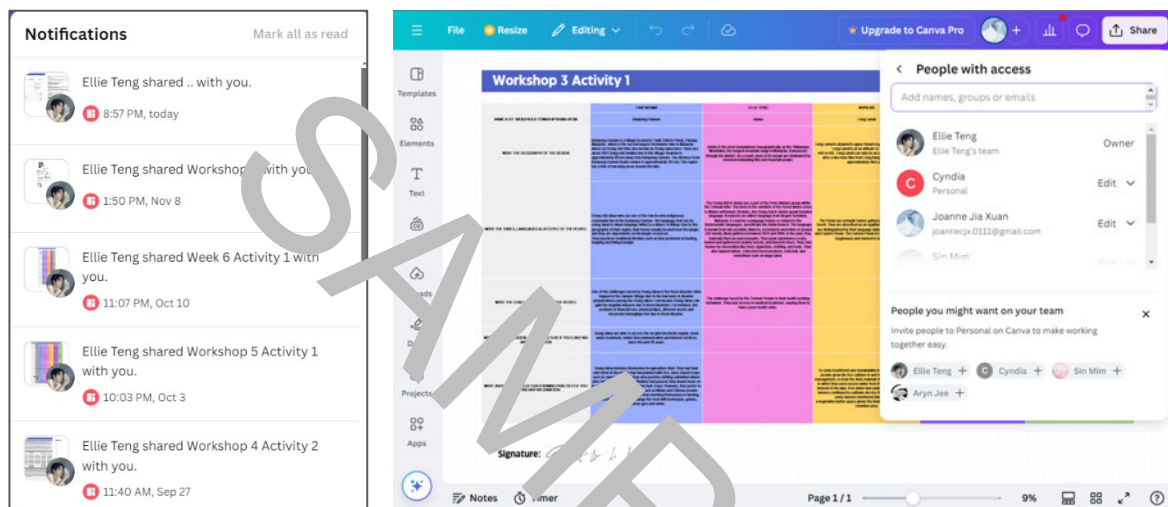


Figure 5. Canva Templates Created by Ellie

As shown in Figure 5, Ellie always prepared the Canva templates before our workshop classes. This proactive effort allowed our team members to begin workshop activities in class without any delays caused by waiting for the template to be created.

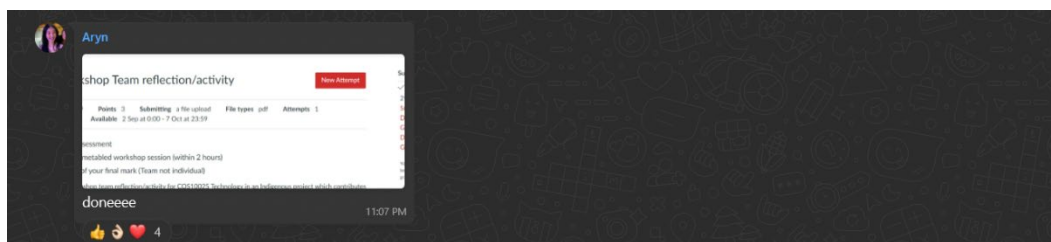


Figure 6. Submission for the Tasks

After our workshops are completed, Aryn would perform final checking to ensure the completed tasks had no mistakes before submission. She then submitted the tasks and shared screenshots in the WhatsApp group to inform us about the submission (see Figure 6).

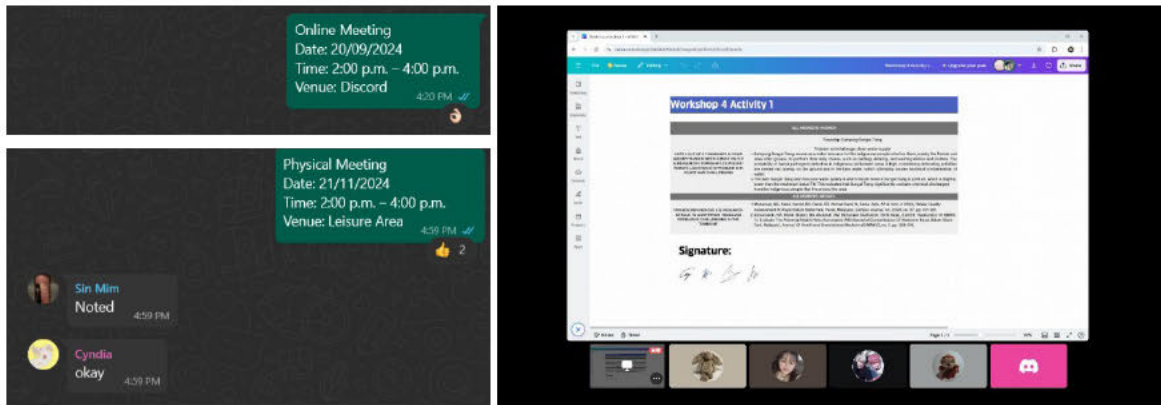


Figure 7. Meeting Announcements & Online Meeting for Workshop 4

Throughout the course of the project, several meetings were held. I organized and announced the meeting dates and venues in the WhatsApp. For virtual meetings, I created a Discord group to facilitate our discussions.

Figure 8. Meeting Minutes about Week 4 Meeting

After each weekly meeting, Cyndia documented the meeting details in the form of meeting minutes. She ensured that all important discussions and decisions were well-recorded for future reference.

Week 5 workshop Team reflection/activity	7 Oct by 23:59	3 / 3	📧 📅
Workshop Team reflection/activity			
Week 6 workshop Team reflection/activity	14 Oct by 23:59	3 / 3	📧 📅
Workshop Team reflection/activity			
Week 7 workshop Team reflection/activity	28 Oct by 23:59	3 / 3	📧 📅 📧 1
Workshop Team reflection/activity			

Figure 9. Full Marks for Weekly Workshop Tasks

This strong collaboration and clear division of responsibilities ensured that all our weekly tasks were completed in high quality, meeting the requirements and earning us full marks (see Figure 9).

← Describe the group work strategies/processes that did not work

2.1.2 Group Work Strategies/Processes that Did Not Work

While every team member performed well in fulfilling their respective roles, we faced challenges with meeting arrangements. Originally, we planned to hold physical meetings as they are generally more effective for communication and reaching solutions. However, the long distance between Aryn's house and the campus made it difficult for her to attend in person. Moreover, Cyndia's busy after-school schedule made it hard to find a time that everyone was available for physical meetings.

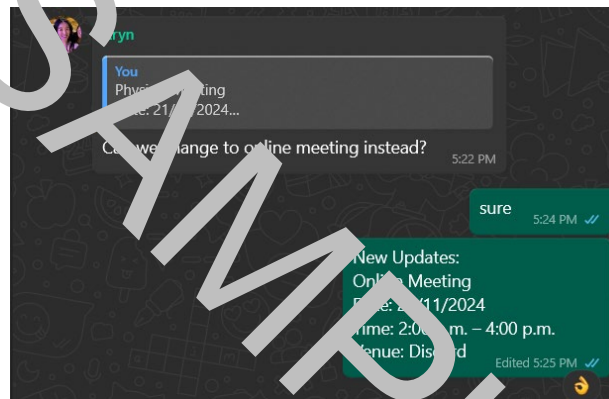


Figure 10. Switching Physical Meeting into Online Meeting

Therefore, we came out with an alternative solution by switching all meetings into virtual ones. Although this resolved the scheduling conflicts, it was less effective than physical meetings. During online meetings, some members occasionally experienced connectivity issues, causing them to have laggy audio when sharing their ideas. This often required them to repeat their points, which disrupted the flow of the discussions. Members might also become less immerse into the discussion through virtual meetings, further impacting the productivity. These challenges affected the efficiency of reaching a solution to the problems discussed.

Describe what could be improved on next time you work in a group.

2.1.3 Recommendations for Future Group Work

Although our group has put in a lot of efforts and performed a great teamwork in this project, certain areas can be improved. The group mainly used WhatsApp and Discord to communicate. This worked well for general communication, but I realized that the features of Microsoft Teams, such as its ability to manage uploaded files, meetings and calendar systematically, made it a more effective tool for future group projects. It offers the ability to create channels for different topics, which can avoid other irrelevant discussions and keep conversations focused. Files uploaded in Teams are also easily accessible, allowing team members to easily find and access the up-to-date versions.

Besides that, I plan to be more proactive in leading discussions in future group work. I can set clear objectives for each meeting and take more initiative to guide the discussions, keeping the conversation focused on key points and avoiding unnecessary side tracks, thus ensuring a more efficient workflow.

Describe an event/action (add evidences) in your team (i.e., not just from you) that you think was outstanding with respect to each of:

- Team organisation
- Meetings
- Delivery of the project design ideas
- Delivery of the Innovation concept
- Delivery of the final presentation

2.1.4 An Outstanding Event

Among the tasks completed in this

remarkable. The task started with a meeting to discuss the distribution of presentation parts based on the design justification score for each of our designs. The presentation tasks were divided as follows: Sin Mim and I were in charge of presenting the best design ideas, Cyndia and Aryn covered their designs as the least design ideas and Ellie worked on the introduction and conclusion of the presentation, which included the problem statement and the design justification table.

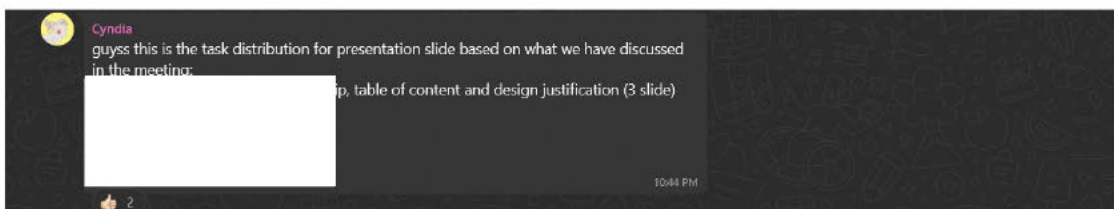


Figure 11. Distributed Tasks for Presentation during Meeting in Week 10

Ellie then created and shared the link to PowerPoint slides with the team (see Figure 12). I was responsible for designing the PowerPoint template to ensure a consistent look across all slides with visual cues that effectively conveyed our design information during the presentation. Each team member then worked on their respective sections of the slides.

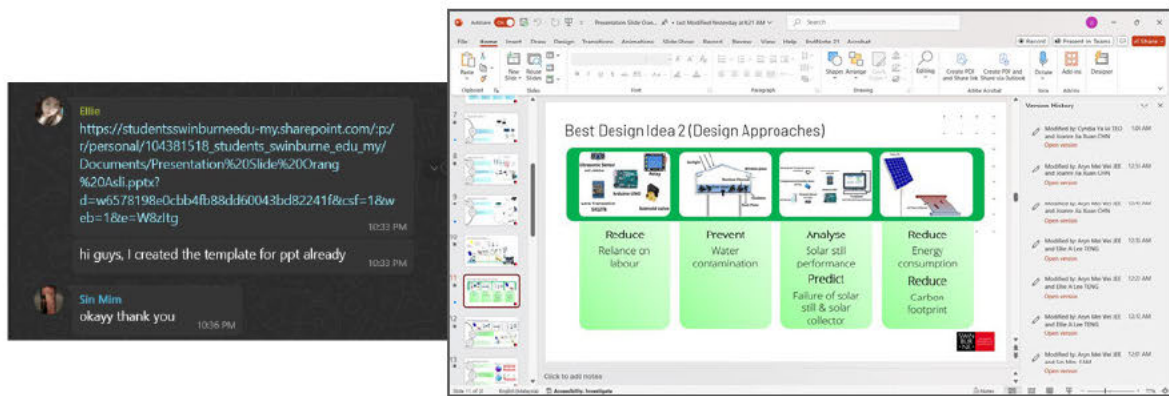


Figure 12. Link to Presentation Slides & Proof of Team Members' Contribution

Throughout the process of populating the presentation slides with our content, teams members continuously communicated to ensure that the information was clear and accurate. For example, Ellie had asked the team members to review her section once she had completed it (see Figure 13)

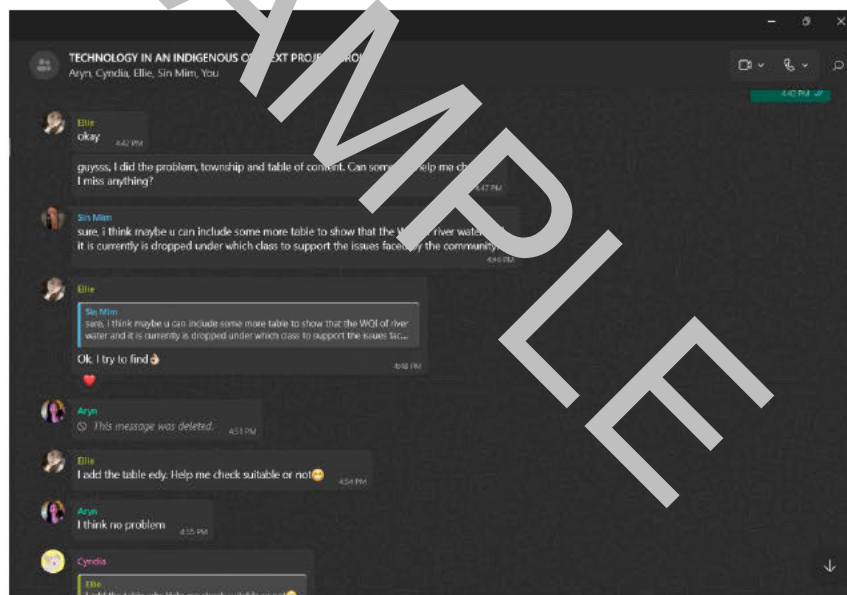


Figure 13. Communication between Team Members on Reviewing Slides

Before the presentation day, several rehearsal meetings were held to ensure everything would run smoothly. During these meetings, each member helped each other with their scripts, ensuring smooth transitions between different speakers. Fortunately, we were able to have a physical meeting as the final rehearsal, which allowed us to simulate the real presentation environment and polish our final delivery.

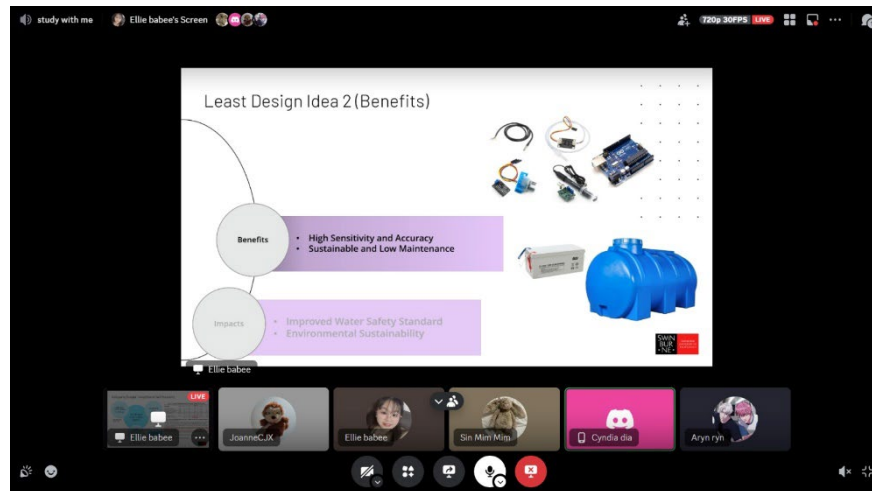


Figure 14. A Scene from Our Presentation Rehearsal

The final presentation was a success. Every team member delivered their ideas thoroughly without any complications. The collaboration within the team, from the design of the slides to the rehearsals, played a crucial role in achieving a cohesive and well-executed presentation.

This section is 1-2 pages. In this section, refer to events and work (week 2 – week 11 group activities) undertaken specifically by you. add evidences from weekly workshop team activities, weekly seminar reflections or assessments

The summary of an individual reflection is well-articulated with project tasks (all four phases), conclusion, and recommendation. Clearly shared learning experiences on contributions to the group.

2.2 Individual Work Reflection

This section discusses my personal reflection on the work I contributed to the project.

2.2.1 Project Tasks

Describe your tasks in the group project in each phase of the project (add evidences)

Phase 1: Identifying Township and the Team Problem

During Week 3, each of us was required to conduct research on five different indigenous townships within Malaysia and identify the challenges faced by the local communities. I chose to investigate Kampung Sungai Tiang, a remote township located in the Royal Belum State Park, Perak. Through my research, I gained a deeper understanding of their living environment, culture and the challenges faced by the Orang Asli community. The challenges I identified were aligned with the scope provided, which included the lack of 24 hours electricity supply, inadequate access to clean water and limited availability to healthcare services.

Workshop 3 Activity 1					
NAME A 10 INDIGENOUS TOWNSHIP IN MALAYSIA	1. PHASE 1: IDENTIFYING TOWNSHIP AND THE TEAM PROBLEM	2. PHASE 2: IDENTIFYING TOWNSHIP AND THE TEAM PROBLEM	3. PHASE 3: IDENTIFYING TOWNSHIP AND THE TEAM PROBLEM	4. PHASE 4: IDENTIFYING TOWNSHIP AND THE TEAM PROBLEM	5. PHASE 5: IDENTIFYING TOWNSHIP AND THE TEAM PROBLEM
	Kampung Sungai Tiang	Kampung Sungai Tiang	Kampung Sungai Tiang	Kampung Sungai Tiang	Kampung Sungai Tiang
WRITE THE GEOGRAPHY OF THE REGION	Kampung Sungai Tiang is a village located in Taik Chai in the Taiping District, Perak. It is a small village with a population of about 100 people. The village is situated in a hilly area with a river flowing through it. The village is surrounded by forest and is a part of the Royal Belum State Park.	Kampung Sungai Tiang is a village located in Taik Chai in the Taiping District, Perak. It is a small village with a population of about 100 people. The village is situated in a hilly area with a river flowing through it. The village is surrounded by forest and is a part of the Royal Belum State Park.	Kampung Sungai Tiang is a village located in Taik Chai in the Taiping District, Perak. It is a small village with a population of about 100 people. The village is situated in a hilly area with a river flowing through it. The village is surrounded by forest and is a part of the Royal Belum State Park.	Kampung Sungai Tiang is a village located in Taik Chai in the Taiping District, Perak. It is a small village with a population of about 100 people. The village is situated in a hilly area with a river flowing through it. The village is surrounded by forest and is a part of the Royal Belum State Park.	Kampung Sungai Tiang is a village located in Taik Chai in the Taiping District, Perak. It is a small village with a population of about 100 people. The village is situated in a hilly area with a river flowing through it. The village is surrounded by forest and is a part of the Royal Belum State Park.
WRITE THE THREE LARGEST SUBSETS OF THE TOWNSHIP	Orang Asli (Melanau) are one of the indigenous groups living in the region. They are known for their traditional way of life and their connection to the forest. They are also known for their traditional crafts and their knowledge of the forest.	Orang Asli (Melanau) are one of the indigenous groups living in the region. They are known for their traditional way of life and their connection to the forest. They are also known for their traditional crafts and their knowledge of the forest.	Orang Asli (Melanau) are one of the indigenous groups living in the region. They are known for their traditional way of life and their connection to the forest. They are also known for their traditional crafts and their knowledge of the forest.	Orang Asli (Melanau) are one of the indigenous groups living in the region. They are known for their traditional way of life and their connection to the forest. They are also known for their traditional crafts and their knowledge of the forest.	Orang Asli (Melanau) are one of the indigenous groups living in the region. They are known for their traditional way of life and their connection to the forest. They are also known for their traditional crafts and their knowledge of the forest.
WRITE THE CHALLENGES FACED BY THE PEOPLE	One of the challenges faced by the Orang Asli is the lack of access to basic services such as electricity, clean water, and healthcare. They also face challenges related to land rights and the impact of deforestation on their traditional way of life.	One of the challenges faced by the Orang Asli is the lack of access to basic services such as electricity, clean water, and healthcare. They also face challenges related to land rights and the impact of deforestation on their traditional way of life.	One of the challenges faced by the Orang Asli is the lack of access to basic services such as electricity, clean water, and healthcare. They also face challenges related to land rights and the impact of deforestation on their traditional way of life.	One of the challenges faced by the Orang Asli is the lack of access to basic services such as electricity, clean water, and healthcare. They also face challenges related to land rights and the impact of deforestation on their traditional way of life.	One of the challenges faced by the Orang Asli is the lack of access to basic services such as electricity, clean water, and healthcare. They also face challenges related to land rights and the impact of deforestation on their traditional way of life.
WRITE ABOUT THE SOCIAL INFRASTRUCTURE IF YOU GAINED ANY INFORMATION	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.
WRITE ABOUT THE INFRASTRUCTURE IF YOU GAINED ANY INFORMATION	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.	Orang Asli are able to access the nearest electricity supply, clean water treatment, and healthcare services through the government's support.

Figure 15. Indigenous Townships Research done by the Group

In Week 4, we discussed and voted on which township the group would proceed with for the project. The township I investigated was chosen, and based on the challenges I have identified, we decided to focus on providing design ideas on solving the issue of clean water access. This decision was then included in Workshop 4. To support the identified problem, I was assigned the task of finding additional articles to provide supporting evidence. The first article presented in Workshop 4 Activity 1 (see Figure 16) was one of my contributions.

Workshop 4 Activity 1	
ALL MEMBERS ANSWER	
VOTE 1 OUT OF 5 TOWNSHIPS IN YOUR GROUP FOUND IN WEEK 3 BASE ON: 1) IT IS INDIGENOUS TOWNSHIP LOCATED AT REMOTE LOCATION 2) A PROBLEM IS IN SCOPE AND CHALLENGING	Township: Kampung Sungai Tiang Problem and challenge: clean water supply - Kampung Sungai Tiang serves as a water resource for the indigenous people who live there, mainly the Temiar and Jahai ethnic groups, to perform their daily chores, such as bathing, drinking, and washing dishes and clothes. The probability of faecal pathogens detection in indigenous settlement areas is high, considering defecating activities are carried out openly on the ground and in the lake water, which ultimately causes bacterial contamination of water. - The river Sungai Tiang also has poor water quality in which the pH level in Sungai Tiang is pH 6.22, which is slightly lower than the neutral pH value 7.0. This indicates that Sungai Tiang significantly contains chemical discharged from the indigenous people that live around the area.
ALL MEMBERS ANSWER	
PROVIDE REFERENCES E.G. RESEARCH ARTICLE TO CONFIRM UN-RESOLVED PROBLEM IS CHALLENGING IN THE TOWNSHIP	1. Mohamad, ND, Abdul Hamid, NS, Ramli, RR, Ahmad Kamil, N, Abdul Adis, AA & Amir, A 2023, 'Water Quality Assessment At Royal Belum State Park, Perak, Malaysia', <i>Larhyss Journal</i>, vol. 2024, no. 57, pp. 101-120. 2. Kamaruddin, NA, Mohd Shahid, NS, Abdullah, AM, Mohamad Shaifuddin, SN & Rajan, S 2023, 'Application Of QMRA To Evaluate The Potential Health Risks Associated With Bacterial Contamination Of Waters In Royal Belum State Park, Malaysia', <i>Journal Of Health and Translational Medicine (JUMMEC)</i>, no. 2, pp. 309-316.

Figure 16. Township Voted based on Workshop 3 Activity 1

Phase 2: Developing Design Idea

After identifying the problem, we brainstormed various design ideas based on the requirements outlined for the project. The articles I reviewed and listed in Workshop 5 (see Figure 17) guided me to come out with my design idea. My initial design concept combined a wireless sensor network, solar still and water pump controlling system (see Figure 18). This concept integrated emerging technologies including solar water purification, Internet of Things (IoT), smart sensors and LoRa technologies. The proposed idea aimed to provide comprehensive water quality monitoring across different river regions and purify the water using solar still.

- López Ramirez, GA & Aragón-Zavala, A 2023, "Wireless sensor networks for water quality monitoring: a comprehensive review", *IEEE Access*, vol. 11, pp. 95120-95142.
- Razman, NA, Wan Ismail, WZ, Abd Razak, MH, Ismail, I & Jamaludin, J 2022, "Design and analysis of water quality monitoring and filtration system for different types of water in Malaysia", *International Journal of Environmental Science and Technology*, vol. 20, pp. 3789-3800.
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- Abhishek, S, Rahul, M, Anjana, E, Isima, S & Kumar, AS 2018, "Water purification using solar thermal and solar PV", *International Conference on Emerging Trends and Innovations in Engineering and Technological Research (ICETIETR)*, pp. 1-4.
- Choubey, S, William, P, Pawar, AB, Jawale, MA, Gupta, K & Parganiha, V 2022, "Intelligent Water Level Monitoring and Water Pump Controlling System using IOT", *2022 3rd International Conference on Electronics and Sustainable Communication Systems (ICESC)*, pp. 423-427.

Figure 17. Reviewed Articles in Workshop 5



Figure 18. Initial Design Drawing from Workshop 5

JOANNE CHIN					
- Analyse - Reduce - Predict	Wireless Sensor Network Device: - sensor nodes Hardware: - multiparameter lab-on-chip - microcontroller (ATmega328PA) - Transceiver - LoRaWAN gateway	High accuracy and sensitivity Multiple water quality parameters are taken at different areas of the river, ensuring an accurate water quality index. Wide coverage The sensor nodes are connected together using LoRaWAN, transmitting data across extensive distances. Low power consumption The sensor nodes are designed to operate with low power, which extends the functional lifespan of the sensors. Low Operational Cost The electricity of the water purification process is generated through solar panel. High purification capability The purification system uses UV rays to evaporate river water. The distilled water can be free of bacteria and toxic substances.	Reduced Pollution Solar power is used in the design, ensuring no toxic gases are produced during the operation. Improve Health Conditions of Indigenous Community Clean water supply, which is free of bacteria and toxic substances, can prevent waterborne diseases and improves the health of indigenous community. Real-time Monitor of Water Quality The water quality monitoring data from the sensor nodes is sent to the cloud from time to time. The residents also can monitor the water quality from the electric pump through the integrated LCD display. Reduce Manpower in Water Quality Monitoring The automated water quality monitoring from time to time minimizes the need of manpower in water specimen data collection and analysis.	Access & Equity The operation of the real-time water quality monitoring system does not require the indigenous community to have other knowledge on how to use it. They can just get the purified water from the water reservoir, which is ready to be consumed. Health & Safety The solar purification system can eliminate bacteria and other heavy metals as the water evaporates, making it safe to drink and free from waterborne diseases. Environmental Health The design of the water quality monitoring system which uses long-lasting battery and solar purification system ensures minimal environment impact. It effectively monitor water quality and purify water without producing harmful substances. Sustainable Livelihoods The long-lasting battery for water quality monitoring system, combined with solar energy used for water purification system ensures the long-term stability of clean water access. By depending on renewable energy and durable technology, the design supports sustainable livelihoods.	Sediment Buildup The sediments accumulated from the water purification process need to be removed periodically to maintain the system efficiency and water quality. High Dependence on Weather The purification system which relies mainly on solar power might be less effective during rainy season as insufficient sunlight is provided to generate electric supply.
- Solar Water Filtration - Internet of Things - Smart Sensors 5G Technologies - LoRaWAN	Solar Water Purification Device: - solar still Hardware: - solar panel - glass - basin Water Pump Controlling System Device: - water pump Hardware: - solar panel - pot sensor - ultrasonic sensor - Arduino Uno - LCD display 12V battery - relay module				

Figure 19. Design Criteria for My Initial Design in Workshop 6

However, after considering cost constraints and the broad scope of the proposed system, I decided to narrow the focus. The revised design (see Figure 20) focuses solely on the solar still water purification process, while incorporating emerging technologies such as IoT and LoRa, along with the software ThingSpeak, to enhance the efficiency of the purification process. The proposed system still aligns at least three design approaches: analyse, reduce and predict, and aligns with the guiding principles.

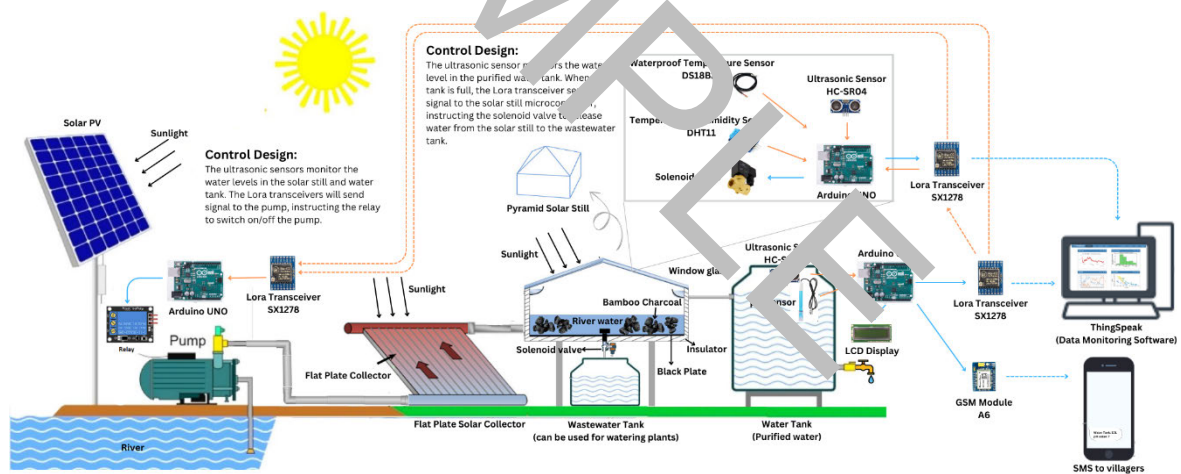


Figure 20. Improved Design in Week 7

Phase 3: Design Justification according to the 6 Guidelines

During this phase, each team member was required to provide evidence to demonstrate how their designs aligned with the design guidelines. The team then evaluated the provided evidence and rated each design ideas based on its adherence to these guidelines. I put forth my utmost effort to provide strong evidence to prove that my design was thoroughly aligned with each of the specified guidelines (see Figure 21). I also carefully assessed the design justification forms of others and provided ratings based on their conformity to the guiding principles.

Table 1 (Individual Work)		Table 2 (Individual Work)					
Design member	Design	Design Justification					
Access and Equity		Joanne Chin	Fan Sin Mim	Ellie Teng	Aryn Jee	Cynda Teo	
AE 1	3	3	3	3	3	3	3
AE 2	3	3	3	3	3	3	2
AE 3	3	3	3	3	3	3	2
Health and Safety		Joanne Chin	Fan Sin Mim	Ellie Teng	Aryn Jee	Cynda Teo	
HS 1	3	3	2	3	3	2	2
HS 2	3	1	1	1	2	1	1
HS 3	3	3	3	3	1	1	1
Environmental health		Joanne Chin	Fan Sin Mim	Ellie Teng	Aryn Jee	Cynda Teo	
EH 1	3	1	1	1	3	1	1
EH 2	3	3	3	3	3	2	2
Appropriateness		Joanne Chin	Fan Sin Mim	Ellie Teng	Aryn Jee	Cynda Teo	
AP 1	3	3	3	3	3	2	2
AP 2	3	3	3	3	3	3	1
Affordability		Joanne Chin	Fan Sin Mim	Ellie Teng	Aryn Jee	Cynda Teo	
AF 1	3	3	3	3	1	1	1
AF 2	3	3	3	2	1	1	1
AF 3	3	2	1	1	1	1	1
Sustainable livelihoods		Joanne Chin	Fan Sin Mim	Ellie Teng	Aryn Jee	Cynda Teo	
SL 1	3	3	3	3	3	3	3
SL 2	3	3	3	3	3	3	3
SL 3	3	3	3	1	3	3	3

Figure 21. My Design Justification Form

After all the designs were rated, Sin Mim and I calculated the average score for each design idea and filled it into the score sheet (see Figure 22). My design ranked first because all sections of the guidelines were supported with solid evidence, proving that my design fulfilled all the guidelines. However, my design did not receive a high score in the health and safety section.

Design member	Design					
Access and Equity						
AE 1	3	2.8	3	3		3
AE 2	3	2.6	3	2.4		2.6
AE 3	3	2.8	2.8	2.6		2.2
Health and Safety						
HS 1	3	2.8	2.8	2.8		2.4
HS 2	1	1	2.8	2.2		1
HS 3	3	3	3	1		1
Environmental health						
EH 1	1	1	3	2.8		1
EH 2	3	2.6	2.8	2.2		2.2
Appropriateness						
AP 1	2.8	2.8	3	3		2.8
AP 2	3	3	3	2.8		1
Affordability						
AF 1	3	2.4	3	1		1
AF 2	3	2.6	2.8	1		1.8
AF 3	2.8	1	3	1		1
Sustainable livelihoods						
SL 1	3	2.8	3	2.8		3
SL 2	3	2.6	3	2.2		2.2
SL 3	3	2	3	2.2		2.2
Average Score	43.6	37.8	47	35		30.4
Rank 1,2,3,4,5	2	3	1	4		5

Figure 22. Average Score for Each Design based on the Six Guidelines

Phase 4: Analysing Design Average Score Sheets and Improvising Design Ideas

Based on the scores given by each member in Figure 22, everyone made improvements to their designs to achieve a higher score. One significant improvement I made to my design was elevating it to higher ground to prevent emergencies such as flooding and animal disruptions. These improvements, made during Phase 4, helped me achieve a full score on the score sheets, resulting in a first-placed ranking, shared with Sin Mim (see Figure 23).

Design member	Design 1C	Design 2C	Design 3C	Design 4C	Design 5C
Access and Equity	6	9	9	8.8	9
AE 1	2	3	3	3	3
AE 2	2	2	2	2.0	2
AE 3	2	3	3	3	3
Health and Safety	6	8.8	9	5.6	5.4
HS 1	3	2.8	3	2.8	2.8
HS 2	3	3	3	2.8	2.6
HS 3	3	3	3	0	0
Environmental health	6	6	6	5	5.6
EH 1	3	3	3	2.6	3
EH 2	3	3	3	2.4	2.8
Appropriateness	6	5.8	6	6	6
AP 1	3	3	3	3	3
AP 2	3	2.8	3	3	3
Affordability	9	5	9	7.8	6
AF 1	3	3	3	2.9	3
AF 2	3	2	3	2.4	2.6
AF 3	3	0	3	2.6	0
Sustainable livelihoods	9	9	9	7.2	8.6
SL 1	3	3	3	2.8	3
SL 2	3	3	3	2	3
SL 3	3	3	3	2.4	2.6
Average Score	46	43.6	46	40.4	40.8
Rank 1,2,3,4,5	2		1	5	4

Figure 23. Final Score for Each Design Justification

The final scores in Figure 23 were then analysed and presented in charts (see Figure 24).

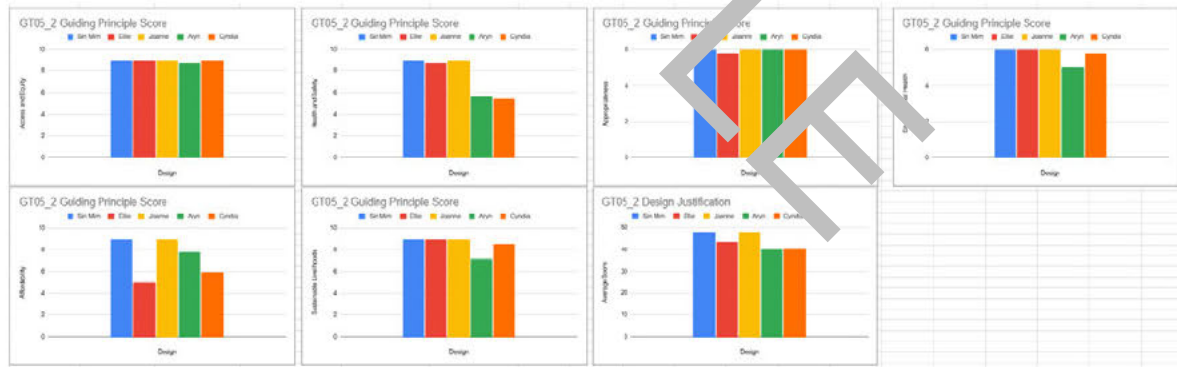


Figure 24. Charts based on Guiding Principles Scores

- Contributions to the group
- o Describe how your efforts contributed to the whole group

o Describe how you were involved in the teamwork environment

2.2.2 Contributions to the Group

During the early stages of the project, the township I investigated in Week 2 was chosen as the problem statement. This choice paved the project direction and provided a clear focus for our designs. Besides that, I was responsible for the design and formatting of the report and presentation slides, ensuring that the information was presented clearly and professionally.

Throughout the project, as the person who was assigned to arrange meetings, I regularly communicate with our team leader, Sin Mim, to ensure our meetings were well-organized and conducted regularly. This helped maintain a steady workflow and high quality of our tasks. This experience made me notice the importance of effective communication among members to ensure that the work is of high quality.

I also actively participated in discussions and assisted team members whenever they needed help. For example, I helped check Part A of the team innovation concept report and made amendments by adding more supporting evidence to strengthen the problem statement. This not only improved the quality of the report but also increased its credibility. Furthermore, for the presentation slides, I helped Ellie out by adding more research content to the problem statement slide. This made her presentation more convincing and easier for the listeners to understand the identified problem. Through active participation, I learnt that by combining different ideas of each team member, a well-thought work can be presented.

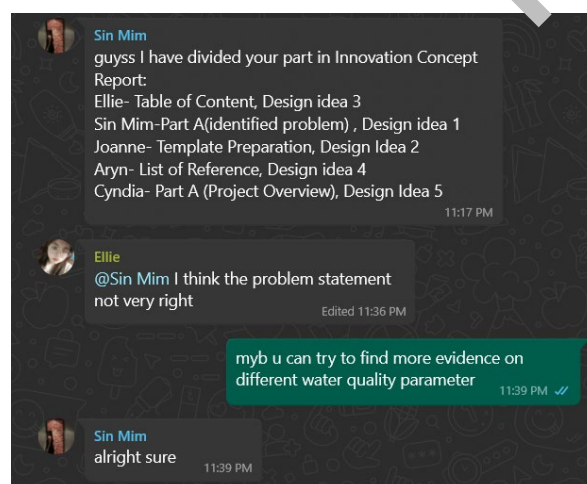


Figure 25. Conversations on Innovation Concept Report

When working with the team, I always remained open to feedback and suggestions on how to improve my design. My team members would often bring different perspectives, which expanded my understanding on how different approach could work. At the same time, I also provided as many suggestions as possible to my team members whenever they had questions or needed assistance. This ensured that everyone was supported and allowed us to resolve any issues together in producing a well completed design.

Conclusion and recommendation

- o Conclude your achievement in accordance with the culturally suitable solution (you can pick either 1 or 2 design ideas that suit well)
- o Recommend how you could further improve your design ideas within a team environment

2.2.3 Conclusion and Recommendation

In conclusion, the design idea I developed combines the solar still water purification system with IoT devices and LoRa technologies, forming a comprehensive automated system. This system operates on clean energy and minimizes the release of harmful substances, making it culturally suitable to be used in Kampung Sungai Tiang. It effectively addresses the issue of insufficient clean water supply while preserving the residents' lifestyle.

Within a team environment, my design idea could be further improved through actively seeking diverse perspectives from team members as everyone may bring up unique perceptions and consider different potential limitations of the system. Thus, this collaborative approach enables the team to together work out a more comprehensive solution for addressing the problem. After the design is improved, the team should have regular meetings to check if my provided design can be ideally implemented.

Part C: Unit Learning Outcomes (ULO)

3.1 ULO 1: Apply relevant knowledge of emerging technologies to a project within an Indigenous context taking into consideration and acknowledging Indigenous histories, worldviews, standpoints, and cultures

Situation:

During Week 4 Workshop 4 Activity 1, our team discussed and selected Kampung Sungai Tiang, a remote Indigenous township one of the townships provided by each member in Workshop 3, as the focus of our project. The township faced a significant problem, which is having limited access of clean water supply. Other than the issue, the selected township also faced challenges such as lack of 24-hour electrical supply and internet connectivity.

Task:

We were assigned the task to design a solution that could address the clean water access problem, while considering the technological and cultural needs of the community. To develop our design, we were required to use at least three emerging technologies from the provided scope and followed at least three out of the six guiding principles.

Action:

In Workshop 5 Activity 1, I provided my design idea, which incorporated the emerging technologies of solar purification system, smart sensors, IoT devices and LoRa technology. These technologies were specifically selected after considering the lack of electricity and internet connectivity in Kampung Sungai Tiang. The solar purification system was



Figure 26. Workshop 5 Activity 1

selected as it could operate without the need for on-grid electricity, ensuring it can operate in remote areas. The smart sensors and IoT devices provide real-time monitoring on the water quality. LoRa technology ensures long distance data transmission, even in areas where there is no internet access. Throughout the development process, I carefully considered the accessibility, affordability and appropriateness of the design to the community. I ensured that the hardware and devices selected were budget-friendly and could be used by the residents without any technical knowledge. The costs of the components were listed in Workshop 5 to further provide further assess on the cost-effectiveness of the design. Moreover, I also considered the cultural practices and lifestyles of the community when designing the system (see Figure 27). Given that the community sometimes would have cultural events such as dancing and worship ceremonies, the system was installed at an originally available space for the water tank, ensuring that the new system would not bring disturbance to their traditions or daily routines.

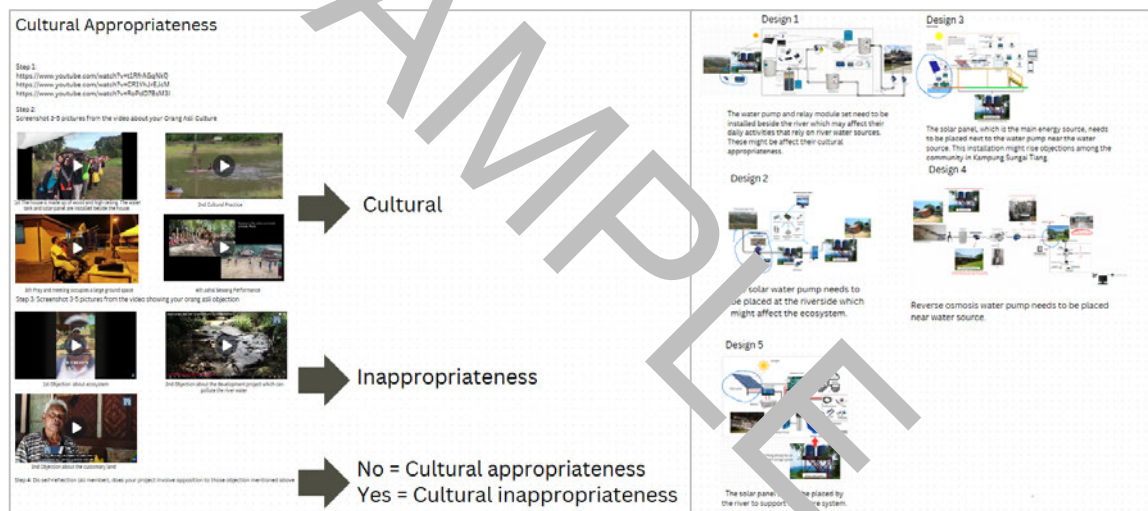


Figure 27. Workshop 9 on Cultural Appropriateness

Response:

By considering all these aspects, the design not only solve issue of insufficient clean water access, but also retain the cultural practices of the indigenous community.

3.2 ULO 2: Function as an effective team member using project management tools and demonstrating professionalism and ethical behaviour

Situation:

Throughout the project, the team needed to perform great teamwork in order to produce high-quality deliverables.

Task:

Each member should be actively involved in group discussions during meetings and workshops and collaborate effectively with each member.

Action:

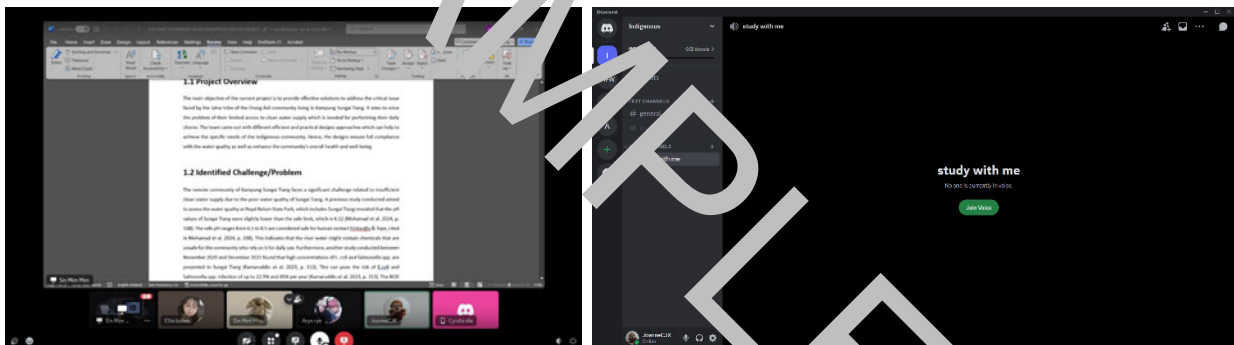


Figure 28. Virtual Meeting for Discussing Innovation Concept Report using Discord

To accomplish this, I organized several meetings every week and announced them through WhatsApp to ensure everyone was updated on progress. I also created a Discord channel for virtual meetings (see Figure 28), which provided a platform for our team to communicate and collaborate remotely. As an effective team member, I actively contributed to the project by adding my work into collaborative templates such as the weekly Canva templates created by Ellie and Word document created by Sin Mim (see Figure 29). I ensured my tasks were delivered on time and to a high standard, demonstrating professionalism by meeting deadlines.

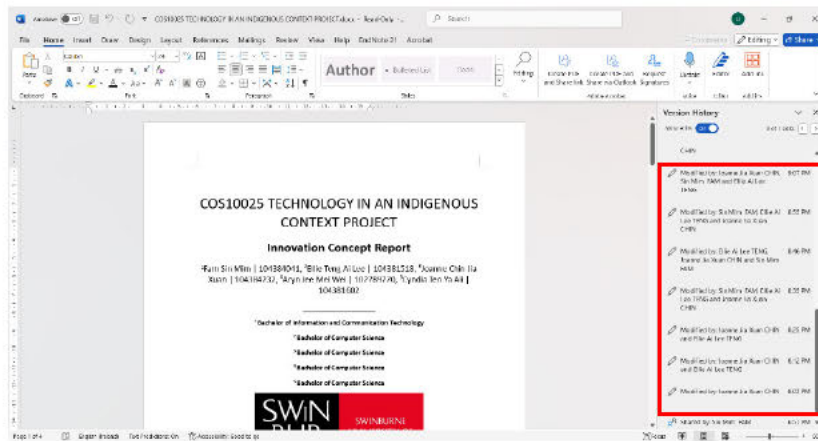


Figure 29. Evidence on My Contribution to Innovation Concept Report

Response:

As a result, all tasks were done on time and had maintained a high standard of quality throughout the project (see Figure 29).

Week 5 workshop Team reflection/activity	23 Oct by 23:59	3 / 3	100%
Week 6 workshop Team reflection/activity	14 Oct by 23:59	3 / 3	100%
Week 7 workshop Team reflection/activity	28 Oct by 23:59	3 / 3	100%
Innovation Concept (Team-based)	7 Nov by 23:59	18 / 20	90%

Figure 30. High Score in Group Tasks

3.3 ULO 3: Communicate within teams, stakeholders using appropriate verbal, written, and technological approaches

Situation:

Communication is essential for the team to come out with the best solutions for the project. As a team, we need to collaborate and discuss ideas to ensure the project met all requirements.

Task:

My task was to communicate clearly with my team members, which involved verbal, written and technological communication approaches.

Action:

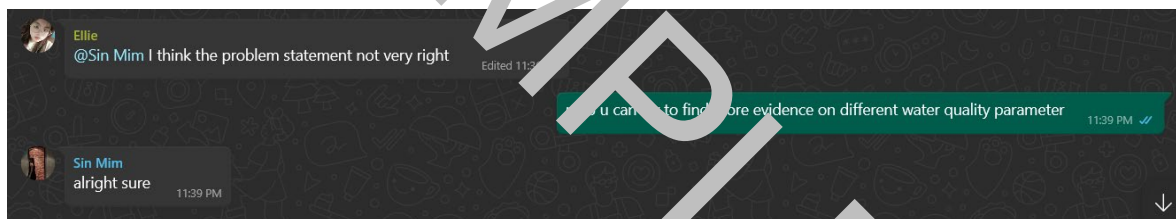


Figure 31. Providing Feedback in WhatsApp

I organized and actively participated in all meetings. During discussions, I voiced my problems or ideas and explained them in full details. This ensured my thoughts were clearly understood by others so that they could provide me with a suitable solution. Besides that, I also provided feedback and replied to questions in the WhatsApp group to help address my teammates issues (see Figure 31). In the reports and weekly tasks, I made sure that my works were written concisely and clearly in the collaborative templates such as Canva (see Figure 32) and Microsoft Word. This allowed my teammates to easily grasp the information and stay aligned with the project's progress. During presentations, I ensured that my attire and body language were professional and appropriate as I recognized that these elements are important form of communication which would affect others' impressions on me.

Action:

- Jasim, MA, Ahmed, OK & Alaiwi, Y 2023, 'Performance of solar stills integrated with PV/thermal solar collectors: A review', <i>NTU Journal for Renewable Energy</i> , vol. 4, no. 1, pp. 97-111.	- Simvula, R, Abo-Ai-Ez, KM & Kahn, MT 2021, 'Maintenance and disposal of the solar panels', <i>International Conference on Electrical, Computer and Energy Technologies (ICECET)</i> , Cape Town, South Africa, 2021, pp. 1-6.
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- Ardenle, F, Beccali, G, Cellura, M & Lo Brano, V 2005, 'Life cycle assessment of a solar thermal collector', <i>Renewable Energy</i> , vol. 30, no. 7, pp. 1031-1054.	- Ardenle, F, Beccali, G, Cellura, M & Lo Brano, V 2005, 'Life cycle assessment of a solar thermal collector', <i>Renewable Energy</i> , vol. 30, no. 7, pp. 1031-1054.
The articles state that the average durability of a solar still and solar PV is around 30 years. The solar collector has an average lifespan of 15 years. Since these two components are the main component in my design, this confirms my design is built to be long-lasting.	
The solar panels only requires regular cleaning of dust to ensure the maximum functionality. The solar still requires regular flush to avoid salt buildup and this is done by the solenoid valve in my design. The solar collector does not need frequent maintenance.	

Figure 33. Evidence provided in Design

Justification Form for Sustainable Livelihood

approach ensures that the system can be used for a long time with only minimal maintenance and replacement required for the main components, ensuring the operational cost for the design. Moreover, the IoT devices used for automated operations ensures the system operate efficiently, consistently functioning to provide continuous clean water to the community.

During Week 9, I also studied on the cultural acceptance of the Orang Asli community in Kampung Sungai Tiang (see Figure 27). Taking those into considerations, my design was planned to be installed at a readily available area which is for originally placing water tanks. This installation ensures that the system brings no interference to the daily routines of the local community and had no conflict with the cultural norms.

Fam Sin Min	104584041	5	2	0	1	0	22	36	16
A	104584042	3	2	0	0	0	26	31	10
C	104584232	7	4	0	1	0	14	20	16
A	102789770	3	9	0	0	0	13	16	11
C	104581602	3	1	0	0	0	10		6

Figure 34. Number of Articles used in Developing my Design Idea up until Week 8

From Week 5 onwards, I have searched and read a lot of articles (see Figure 34) to learn about the implementation of solar water purification, IoT devices and LoRa technology for improving my design. I found out that these emerging technologies are used in different regions of the world, especially in rural areas. Thus, the application of these technologies in my design have been improved weekly to provide a suitable solution specifically for the challenge at Kampung Sungai Tiang.

Response:

By considering the sustainable context and cultural appropriateness of the design, the proposed design might be culturally accepted by the local and promotes long-term sustainability. The implementation of the emerging technologies also aligns the system with the global trends towards renewable energy and sustainable development.

(4673 words)

word count



SAMPLE

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Proofreading of report before submission, make sure no errors in grammar and spelling

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